

# ArchiFlow AI – Cost-Safe Architecture & Pricing Guardrails

## 1. Purpose of This Document

This document defines the **cost-safe system architecture, quota enforcement model, and pricing guardrails** for ArchiFlow AI. It is designed to ensure the platform remains **profitable at USD 20 per user per month** when deployed on Google Cloud.

This document is written for developers, system architects, and product engineers. All requirements in this document are **mandatory unless explicitly stated otherwise**.

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## 2. Business Constraint (Non-Negotiable)

- Subscription price: **USD 20 / user / month**
- Target gross margin: **≥ 45%**
- Architecture must prevent any single user from exceeding their allocated cost budget
- AI usage must be technically enforced, not policy-based

Failure to comply with these constraints will result in financial loss and is unacceptable.

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## 3. High-Level Architecture Overview

### 3.1 Core Design Principle

Google Cloud is used strictly as a **cost-controlled orchestration and hosting layer**, not as an unrestricted AI computation engine.

All AI-related workloads must pass through **quota checks, caching, and async job control** before execution.

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### 3.2 System Components

#### 3.2.1 Frontend (Web Application)

- Technology: React / Next.js (implementation choice flexible)
- Responsibilities:
  - User interaction
  - Diagram editing UI
  - Job submission requests

- Constraints:
  - No direct AI calls
  - No heavy processing
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### 3.2.2 API Gateway (Cost Gatekeeper)

This is the **most critical cost-control component**.

Responsibilities: - Authentication and authorization - Subscription plan validation - Quota validation (before any job execution) - Rate limiting - Request size validation

Rules: - If quota is exceeded, the request must be rejected immediately - No request may proceed to AI processing without passing this layer

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### 3.2.3 Core Backend (Cloud Run)

- Execution model: Serverless (scale-to-zero)
- Responsibilities:
- User management
- Diagram metadata management
- Job orchestration
- Cache lookup and storage
- Export handling

Constraints: - No long-running or expensive AI logic allowed - All heavy work must be delegated to async tasks

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### 3.2.4 Task Queue (Cloud Tasks / PubSub)

All expensive operations must be executed asynchronously.

Handled jobs include: - Document parsing - Diagram generation - AI-based revisions - Diagram exports

Controls: - Concurrency limits - Retry limits (maximum 2 retries) - Execution timeouts

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### 3.2.5 AI Orchestration Layer

This layer governs **all AI usage decisions**.

Responsibilities: - Determine AI provider selection - Enforce quota and usage limits - Check cache before executing AI calls - Enforce retry and timeout rules

Rules: - No AI calls allowed outside this layer - Cache hit must bypass AI execution

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### 3.2.6 Data Storage

- Cloud Storage:
    - Uploaded documents
    - Diagram images
  - Exported files
  - Firestore / Cloud SQL:
    - User plans
    - Usage counters
  - Job states
  - Optional: Redis / Memorystore
  - Hot cache for AI responses
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## 4. Quota Enforcement Model

### 4.1 USD 20 / Month – Pro Plan (Launch Plan)

| Resource              | Monthly Limit |
|-----------------------|---------------|
| Diagrams created      | 20            |
| Document uploads      | 10            |
| AI revisions          | 40            |
| Exports               | 30            |
| Max nodes per diagram | 30            |
| Max layers            | 5             |
| Max file size         | 10 MB         |
| Concurrent AI jobs    | 1             |

These limits must be **enforced in code**.

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### 4.2 Enforcement Flow (Mandatory)

Before any AI job is enqueued: 1. Validate user authentication 2. Validate active subscription 3. Validate remaining quota 4. Validate job size and complexity 5. Validate concurrency limit

If any check fails, the request must terminate immediately.

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## **5. Cost Control Mechanisms**

### **5.1 AI Response Caching**

- Cache key must include:
- Document hash
- Prompt hash
- Persona / mode

Identical inputs must never trigger duplicate AI calls.

Expected impact: - 30-50% AI cost reduction

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### **5.2 Diagram Complexity Limits**

Hard rejection rules: - Node count exceeding limit - Excessive layers - Excessive revisions per diagram

This prevents abusive or accidental overuse.

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### **5.3 Async Execution Controls**

- Max execution time per task
  - Max retries: 2
  - Automatic job termination on timeout
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### **5.4 Budget Guardrails**

Google Cloud budget alerts must be configured: - 50% monthly spend alert - 75% monthly spend alert - 90% automatic throttling mode

At 90% usage: - New AI jobs are blocked - Cache-only responses allowed

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## 6. Cost Model (Per User / Month)

### 6.1 Google Cloud Infrastructure

| Component            | Estimated Cost (USD) |
|----------------------|----------------------|
| Cloud Run            | 1.20                 |
| Database             | 0.40                 |
| Cloud Storage        | 0.30                 |
| Cloud Tasks / PubSub | 0.50                 |
| Network & misc       | 0.30                 |

Subtotal: ~**2.70**

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### 6.2 AI Usage (Controlled)

| Activity                 | Cost (USD) |
|--------------------------|------------|
| Diagram generation       | 2.50       |
| Document parsing         | 1.50       |
| Revisions & explanations | 2.00       |

Subtotal: ~**6.00**

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### 6.3 Operational Buffer

| Item                 | Cost (USD) |
|----------------------|------------|
| Logging & monitoring | 0.50       |
| Retries & failures   | 0.30       |
| Support buffer       | 0.70       |

Subtotal: ~**1.50**

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### 6.4 Total Cost Summary

Total estimated cost per user per month: ~**USD 10.20**

Revenue per user: **USD 20.00** Estimated gross margin: ~**49%**

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## 7. Explicit Non-Launch Features

The following features must **not** be included in the USD 20 plan: - Unlimited AI usage - Real-time AI editing - Continuous AI background analysis - Auto-regeneration loops - Heavy OCR or vision processing

These features are reserved for future enterprise tiers.

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## 8. Launch Readiness Checklist

Before production release, verify: - Quotas enforced at API level - AI calls restricted to orchestration layer - Caching implemented - Budget alerts active - Async jobs enforced - Kill switch tested

Launch must be blocked if any item above is incomplete.

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## 9. Final Directive

This document defines **mandatory architectural and cost constraints**.

Any deviation must be reviewed and approved at product-owner level due to financial risk.